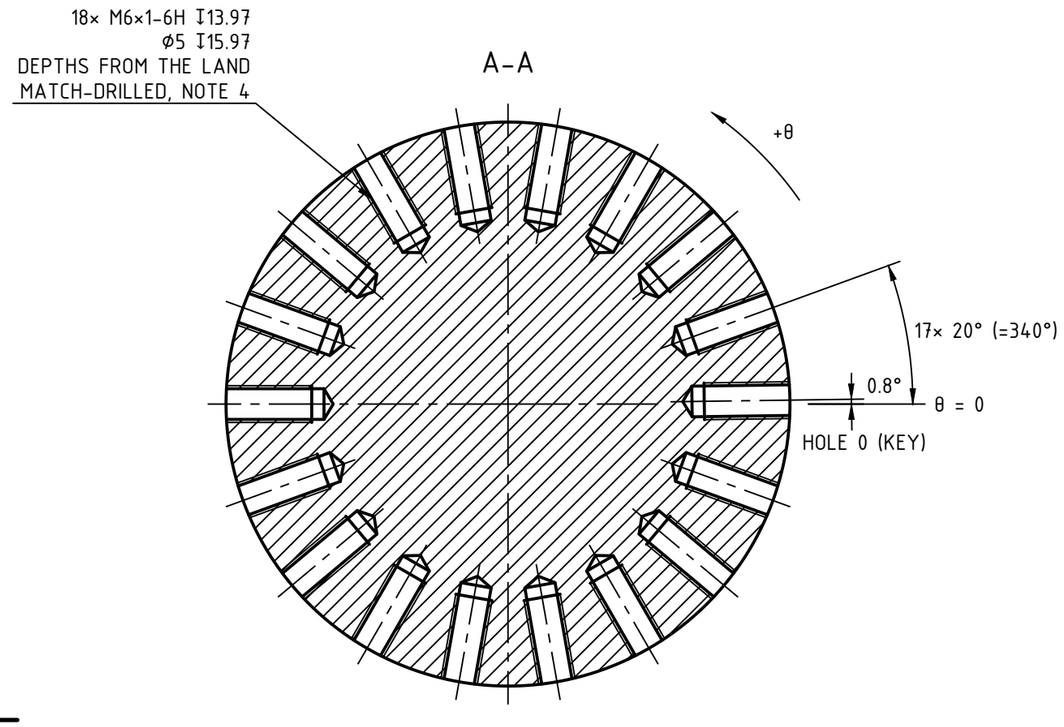

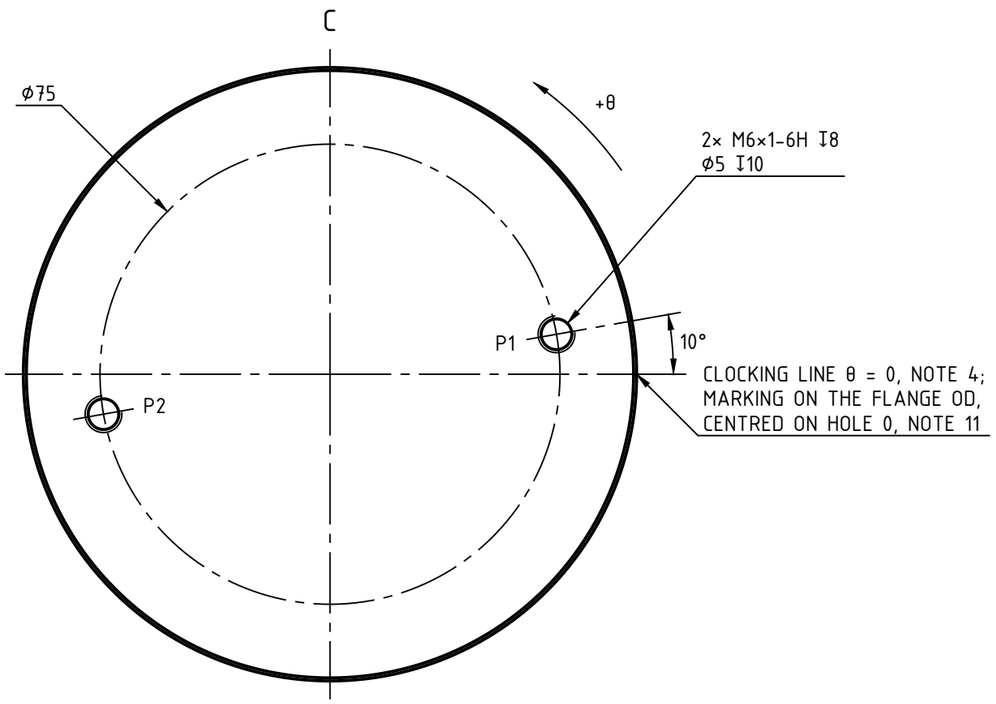

SECTION A–A AND VIEW C ARE SEEN FROM FORWARD: +θ COUNTER-CLOCKWISE, θ = 0 ON +X.
P1 AT θ = 10°, P2 AT 190°; HOLE 0 (θ = 0.8°) IS 9.2° CLOCKWISE OF P1.

NOTES

1. Material: aluminium alloy 6061-T651 round bar to ASTM B211 or ASTM B221 or equal (AMS 4117; B221 extruded bar may come as 6061-T6511), stock ⌀105 × 50 min, grain along the axis. Mill certificate with the heat / lot number required.
2. Bare metal: no anodizing, conversion coat or paint. Texture: land and groove walls Ra 1.6, groove bottoms Ra 0.8, thrust face Ra 1.6, elsewhere Ra 3.2; circumferential lay on the land and grooves. Finish the land, both grooves and face B in one setup.
3. The plug is cut to its own case: land ⌀(A − 0.06) 0/−0.02, both groove bottoms ⌀(A − 5.67) 0/−0.04, where A is the honed size of the mating case's forward seal zone as recorded on SRM100-101 (note 4, R4): the smallest reading in the zone, 92.000 ≤ A ≤ 92.070, the zone held to A +0.030/0 (never over 92.10). ⌀91.94 and ⌀86.33 are drawn for A = 92.000. Record the A used with the serial.
4. Radial holes: made ONLY at assembly by match-drilling, per SRM100-101 ops 110 … 150, this closure seated and clamped in its case (face B on the case forward end face, O-rings out), hole 0 first. Until then θ = 0 is carried by puller P1 (θ = 10°): hole 0 (θ = 0.8°) is 9.2° clockwise of P1 seen from forward (view C). Depth stops from the case OD (the land is covered): full ⌀5 to 20.0 (15.97 from the land), full thread to 18.0 (13.97); bottoming tap through the reamed case holes. The ream entry cone of SRM100-101 op 130 is permitted in each hole (0.5 deep max, ⌀6.40 at the land); full thread is counted below it. Scribe the clocking line θ = 0 across the case end and the flange OD, 0.05 deep max. Case and closure are then a serialised pair: never interchange.
5. The 18 radial tap drills end at r 28.50 in solid metal; the closest pair (holes 0 and 1, 19.2° apart) leave 4.51 between their points, 2.0 min.
6. Undimensioned chamfers 0.5 × 45° (flange edges); groove edges per detail D. Every edge an O-ring passes over burr-free and smooth; tapped-hole entries on the land flush, no raised burr. After machining and again after match-drilling: blow out, wash and degrease (IPA); no chips in the grooves or threads.
7. No port and no pocket: item 6 is solid on the axis, 44 thick, and its forward face is one flat thrust face, broken only by the 2 M6 puller holes (note 10). Do not drill, tap or pocket it beyond this drawing. The motor is fired without a chamber transducer; a transducer needs a new, ported closure (new drawing) and a re-proof.
8. Item 7 (SRM100-107, EPDM full disc ⌀91.8 × 3) is bonded to the aft face only after the proof test (SRM100-000 sheet 3): leave that face as machined, unmarked. IPA degrease, abrade (P180 / Scotch-Brite), IPA wipe, silicone primer SS4004P, RTV106 or equal over the whole face, bondline 0.1 … 0.2, disc OD centred on the land within 0.1; cure 24 h.
9. O-rings 2 × AS568-237, FKM 90A (Parker V0709-90 or equal), item 13; Parker Super O-Lube or DC-55, thin film. Screws 18 × ISO 7380-1 M6×16-010.9, item 16, Loctite 243, 4 … 6 N·m, criss-cross in 3 passes (SRM100-000 sheet 3).
10. Pullers P1 (θ = 10°) and P2 (190°) on ⌀75: 2 × M6×1-6H ↧8, ⌀5 ↧10, machined before match-drilling; for the puller bridge and two draw screws only (tooling). Radial screws out first; never lever or knock the closure out.
11. Mark 'SRM100-103 FWD', the part serial, the pair serial and A on the flange OD, centred on hole 0 (θ 0.8°), a 2 mm gap left in the text where the clocking line θ = 0 crosses: 2 mm characters, laser or dot-peen, 0.05 deep max. No marks on the thrust face, face B, land, grooves or aft face.
12. Inspect 100 %, record with the serial: land ⌀ (3 positions × 2 directions), groove bottom ⌀ and widths, the geometric tolerances, grooves visually at 10×; after match-drilling every radial thread with the M6-6H GO / NO-GO gauge, and its depth.
13. After every firing (SRM100-000 items): replace items 4, 5, 7, 8, 9, 10, 11, 12, 13, 14, all 36 screws (items 15, 16) and the temperature labels; re-use items 1, 2, 6 after inspection (SRM100-101 R10 for the case), and item 3 while the throat is ≤ ⌀23.9 and uncracked. This closure (item 6): remove item 7 and the old RTV with a plastic scraper (no metal tools), clean, and repeat the note 12 checks: no erosion, heat damage or scoring on the land, grooves or aft face, threads in gauge. Any finding: quarantine, engineering decides.

Ra 3.2 (✓)

USED ON: SRM100-000 item 6 (1 off); SRM100-001 item 6

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
6061-T651 bar, ASTM B211/B221 or equal, stock ⌀105 × 50 min	770.5 g	ISO 2768-mK	Ra 3.2	ISO 13715 −0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED – FOR REVIEW	1:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	FORWARD CLOSURE Section A–A (radial tapped holes), view C, notes

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-103	A	2 / 2